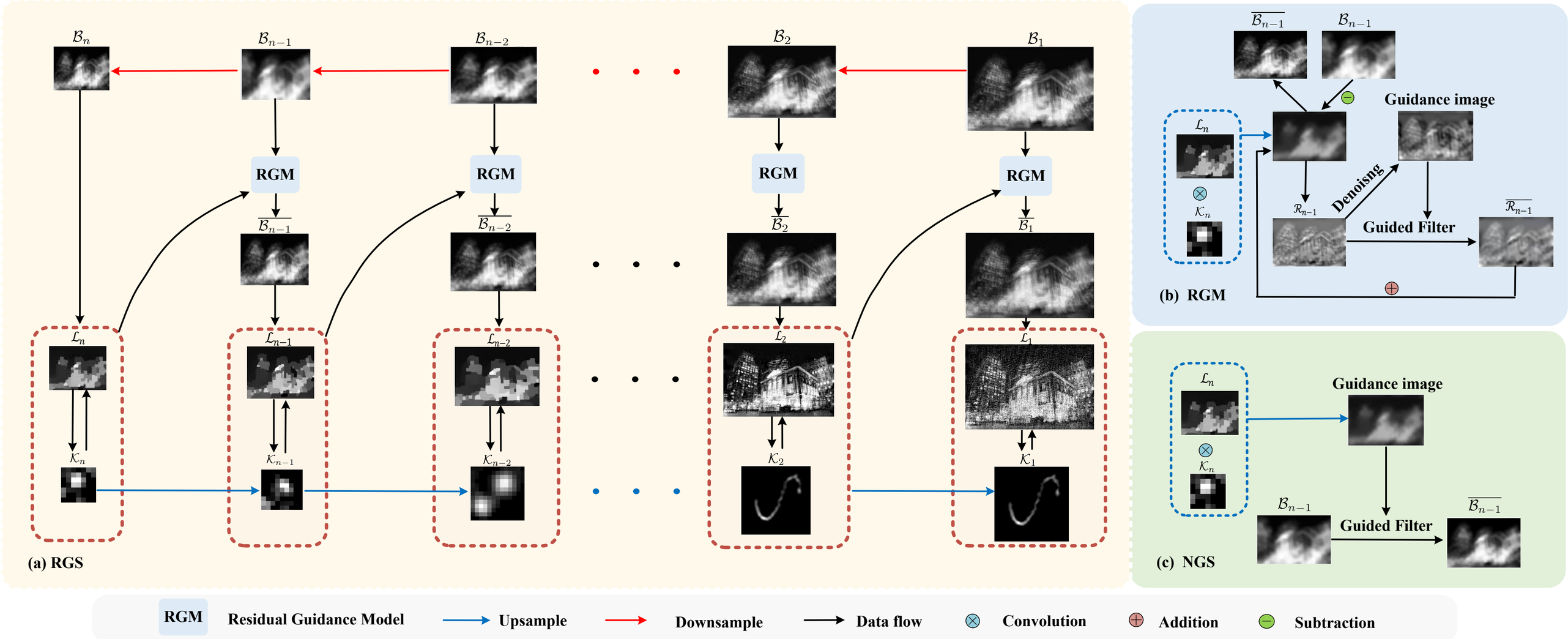
$\mathcal{B}_n$
$\mathcal{B}_{n-1}$
$\mathcal{B}_{n-2}$
$\mathcal{B}_2$
$\mathcal{B}_1$
RGM
$\overline{\mathcal{B}_{n-1}}$
$\overline{\mathcal{B}_{n-2}}$
$\overline{\mathcal{B}_2}$
$\overline{\mathcal{B}_1}$
$\mathcal{L}_n$
$\mathcal{L}_{n-1}$
$\mathcal{L}_{n-2}$
$\mathcal{L}_2$
$\mathcal{L}_1$
$\mathcal{K}_n$
$\mathcal{K}_{n-1}$
$\mathcal{K}_{n-2}$
$\mathcal{K}_2$
$\mathcal{K}_1$
(a) RGS
$\overline{\mathcal{B}_{n-1}}$
$\mathcal{B}_{n-1}$
Guidance image
$\mathcal{L}_n$
$\mathcal{K}_n$
$\mathcal{R}_{n-1}$
Denoising
Guided Filter
$\overline{\mathcal{R}_{n-1}}$
(b) RGM
$\mathcal{L}_n$
Guidance image
$\mathcal{K}_n$
$\mathcal{B}_{n-1}$
Guided Filter
$\overline{\mathcal{B}_{n-1}}$
(c) NGS
RGM
Residual Guidance Model
Upsample
Downsample
Data flow
Convolution
Addition
Subtraction